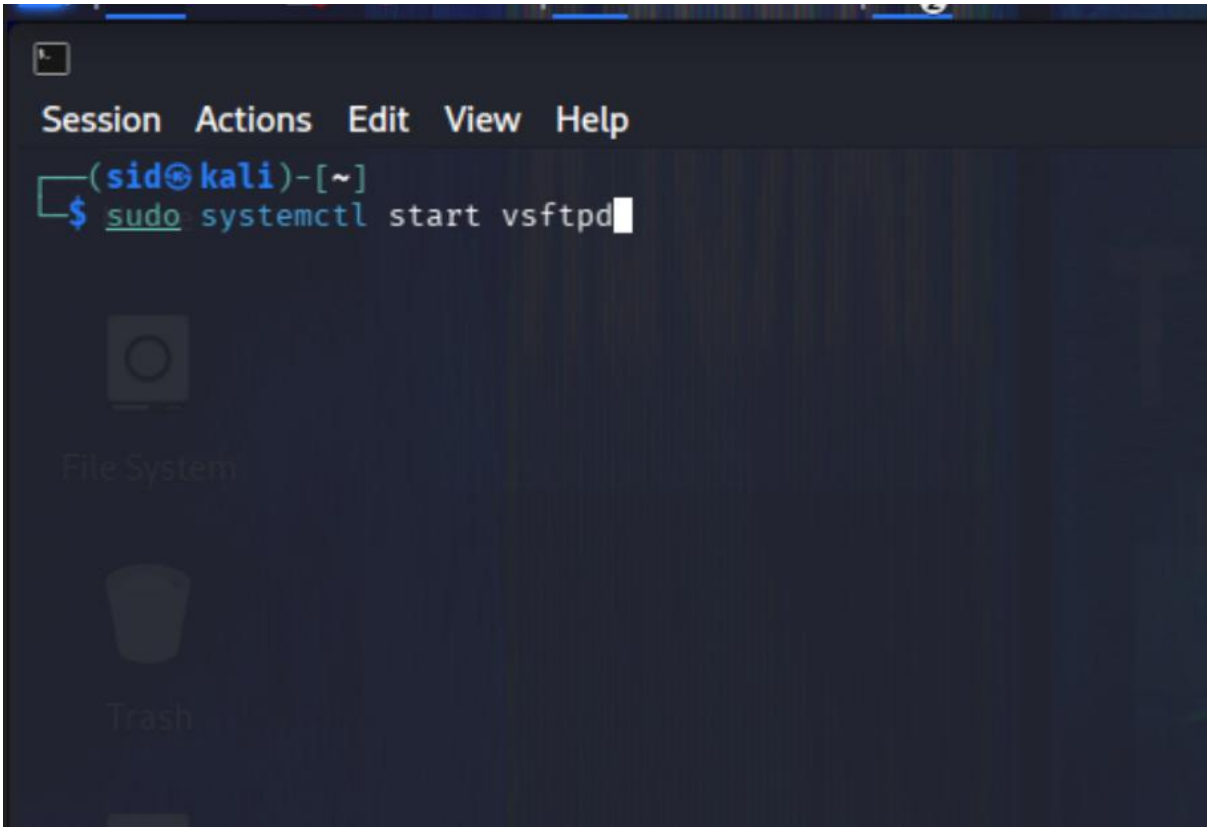
Phase 2 – Intercepted Credentials (Wireshark)

-Lt Siddhant Khajuria

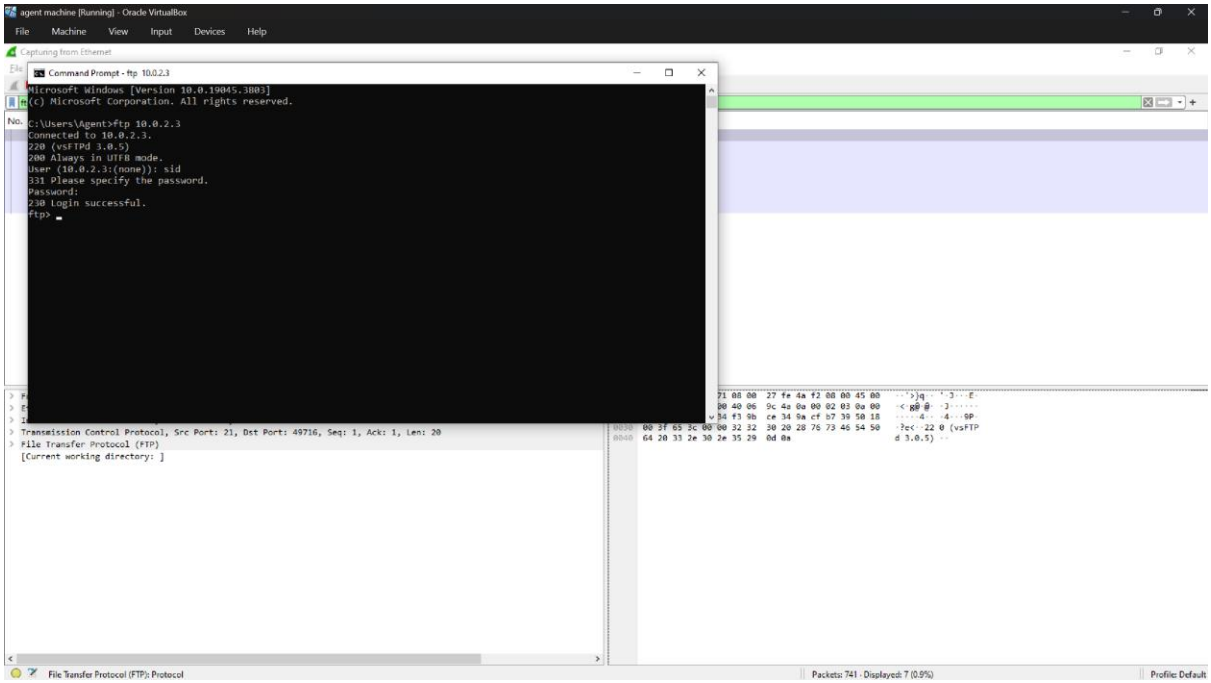
1. **FTP LOGIN**

We assume that an ftp server already exists on the target pc,

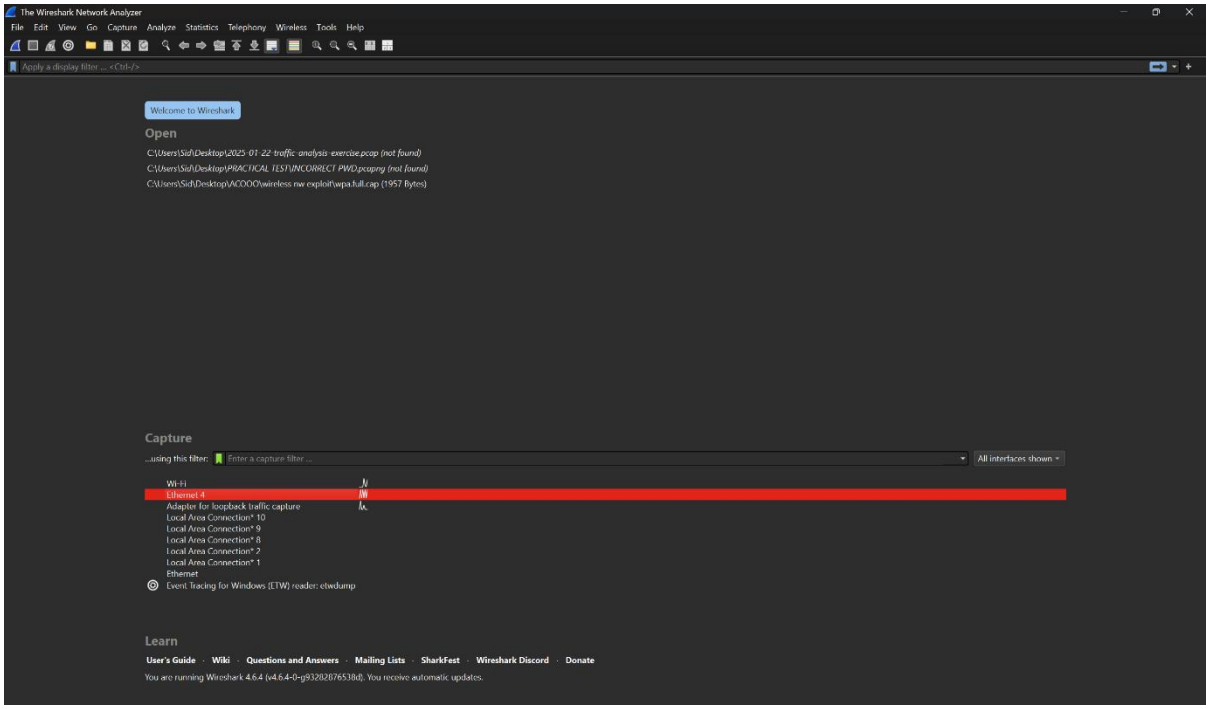
- Assuming the ftp service is started via –



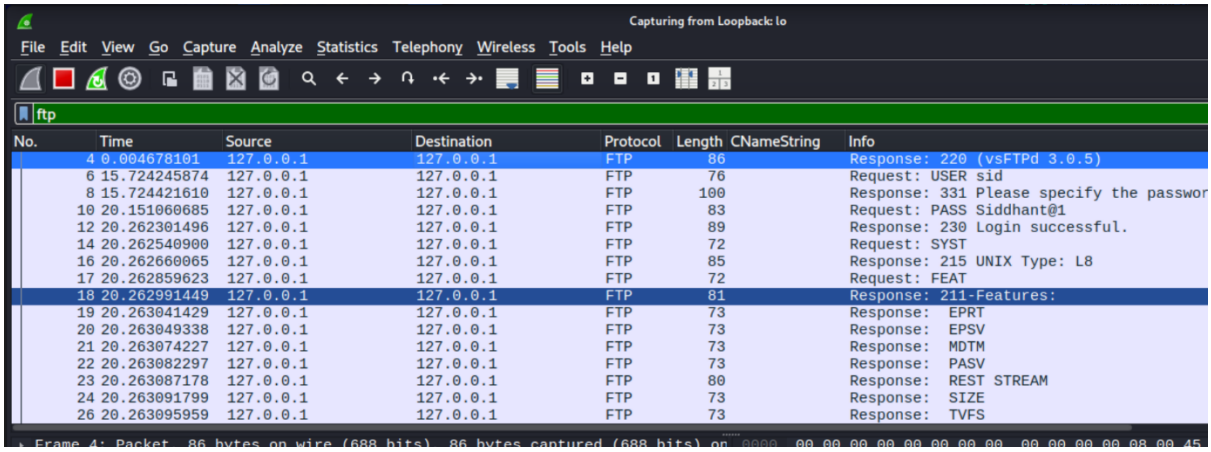
- Victim logs in to his ftp server



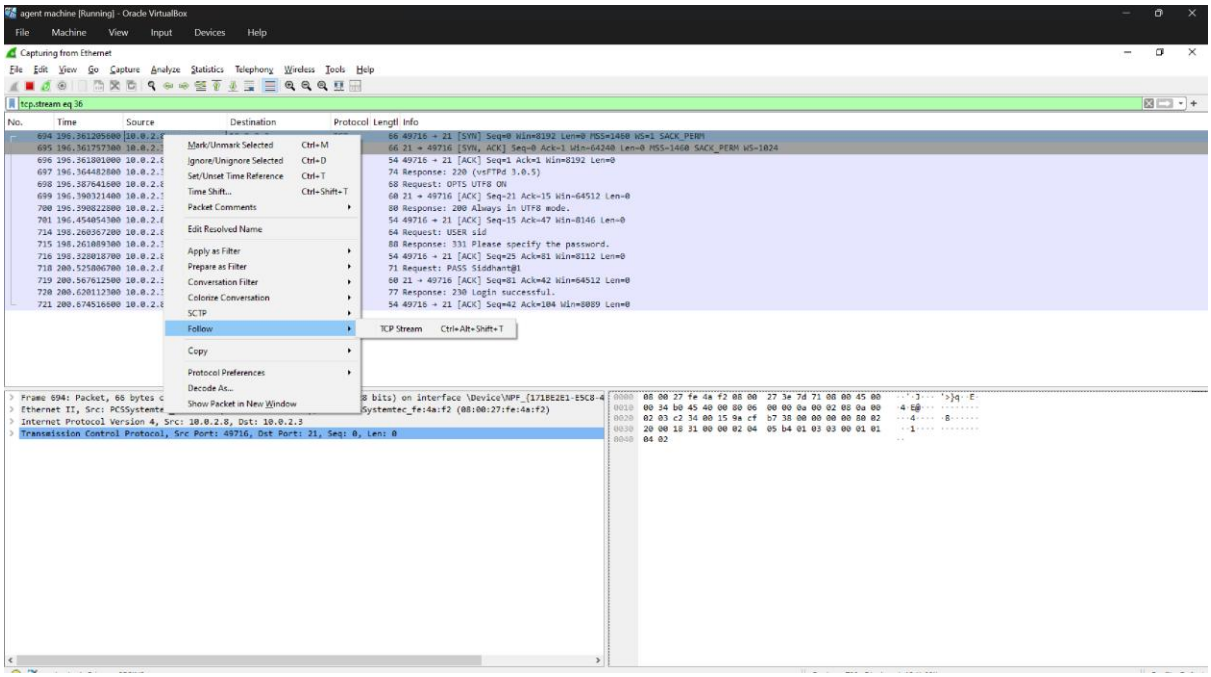
- We open wireshark & select adapter on same org network



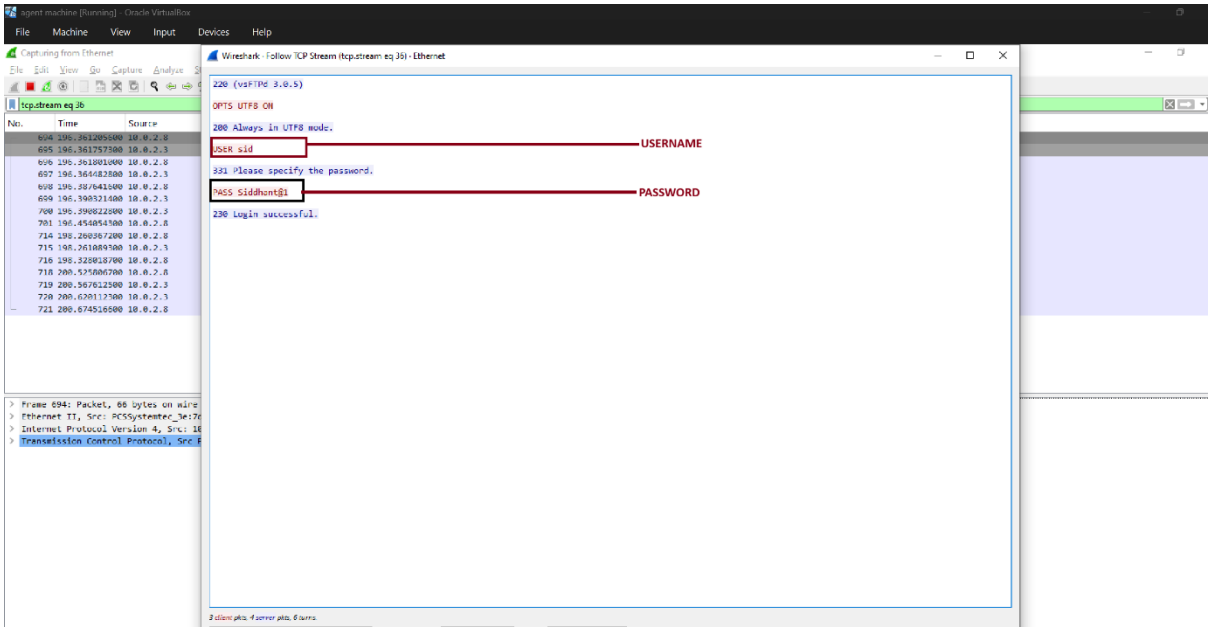
- We filter logs by using “ftp” filter and see ongoing traffic



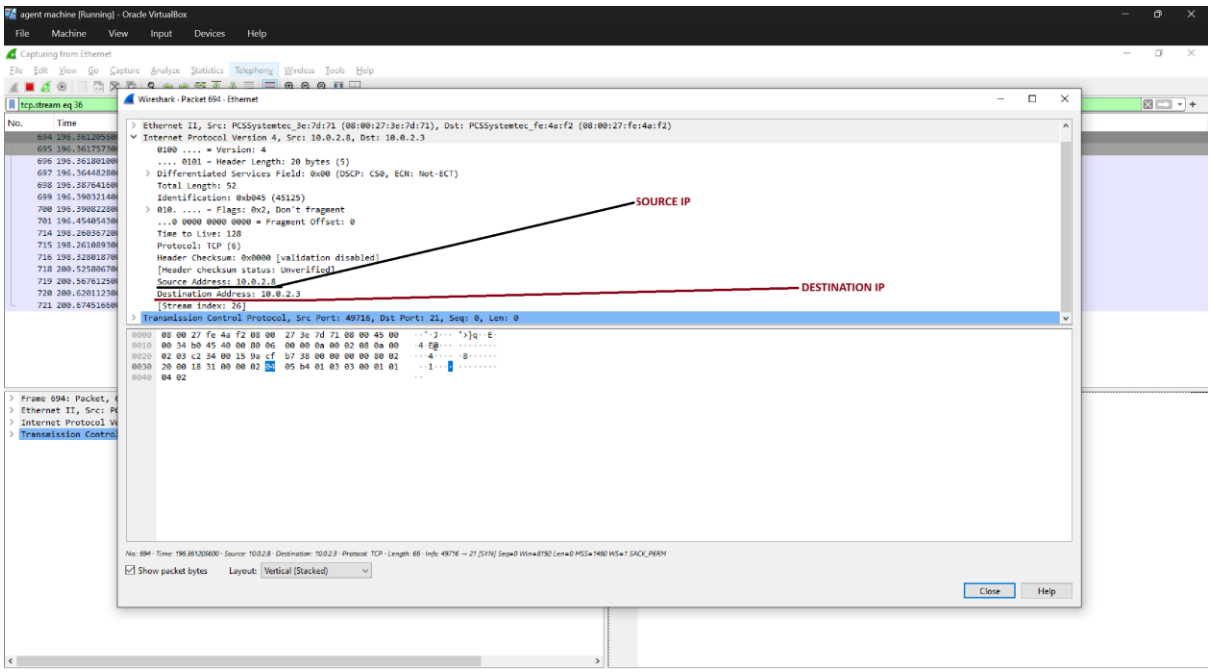
- We follow TCP Stream of ACK packet



- We can see the unencrypted ftp credentials –



- We can also see source and destination IP



2. Why encryption absence is dangerous (FTP)

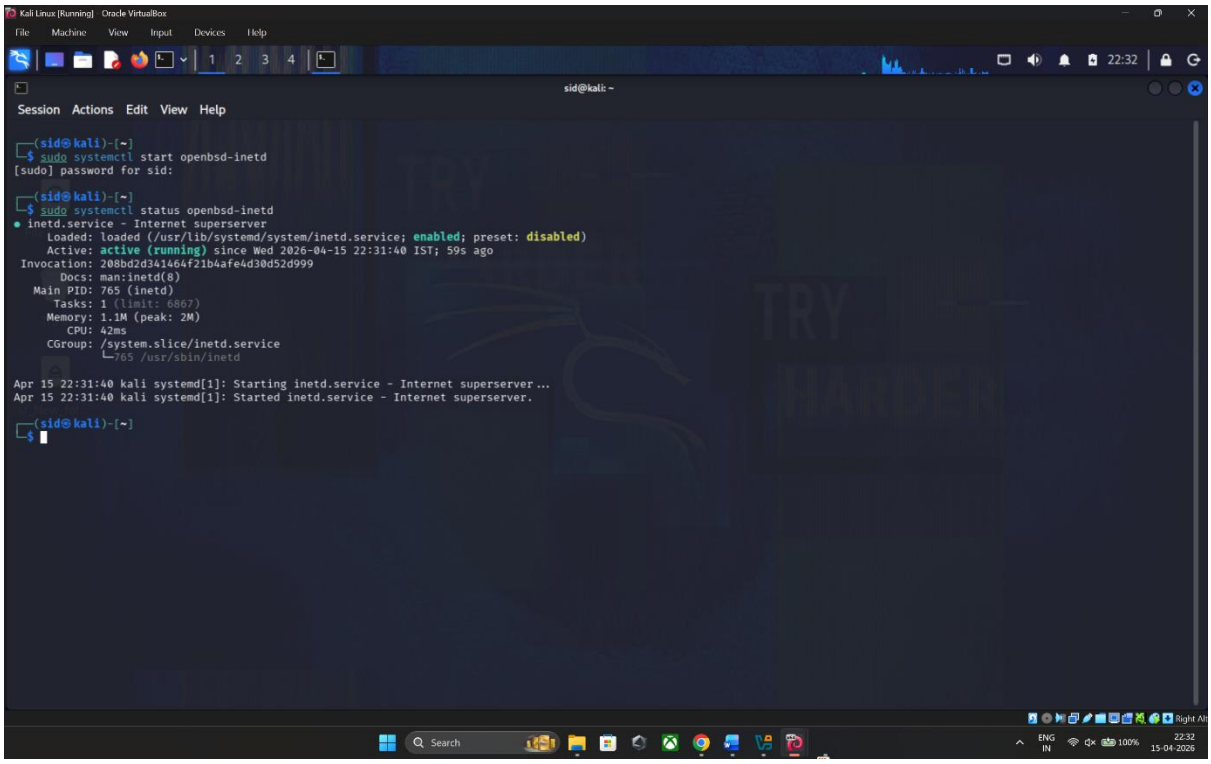
- In a secure protocol like SSH, your password is scrambled before it even hits the network cable. In FTP, it is sent in Plaintext.
- FTP uses two separate channels: one for commands (Port 21) and one for data (Port 20). Neither is encrypted.
- If you use FTP to transfer a sensitive spreadsheet those files are sent across the wire bit-by-bit in an unencrypted stream.
- The lack of encryption and modern authentication in old FTP versions allows for FTP Bounce Attacks.

TELNET

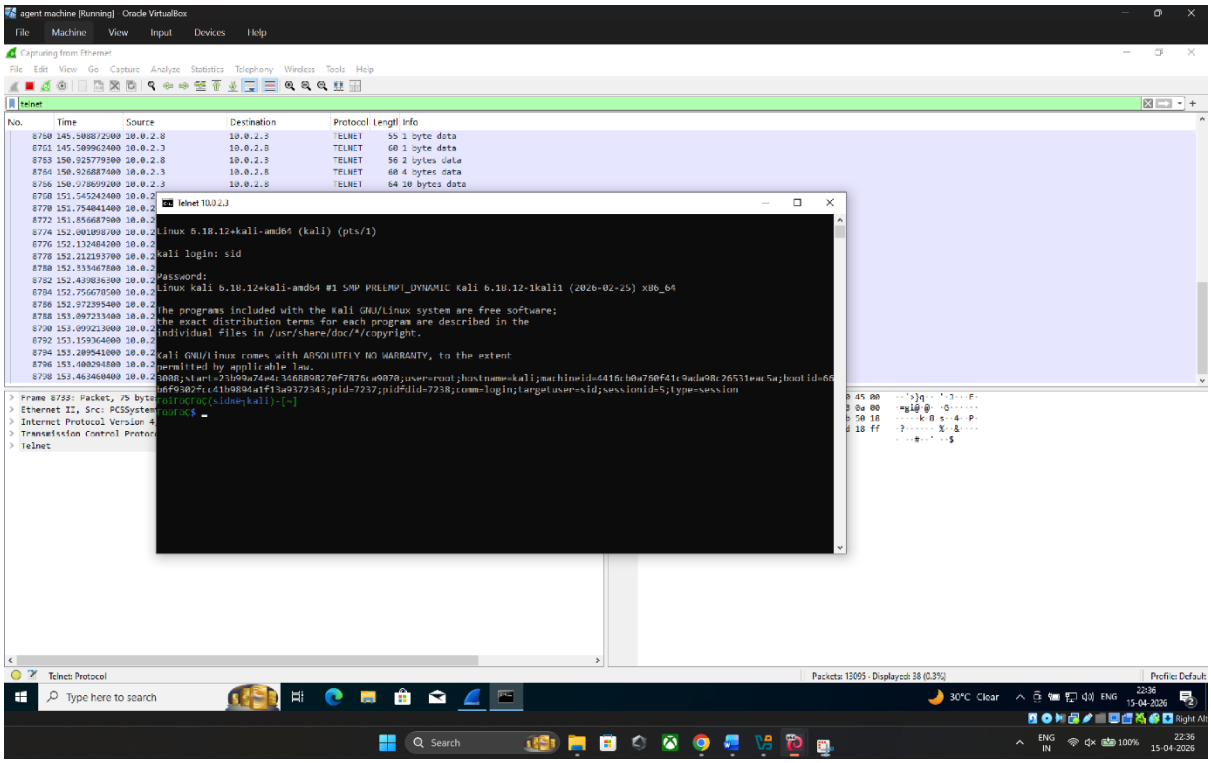
- **Telnet Login**

  Capture username/password sent in plaintext.

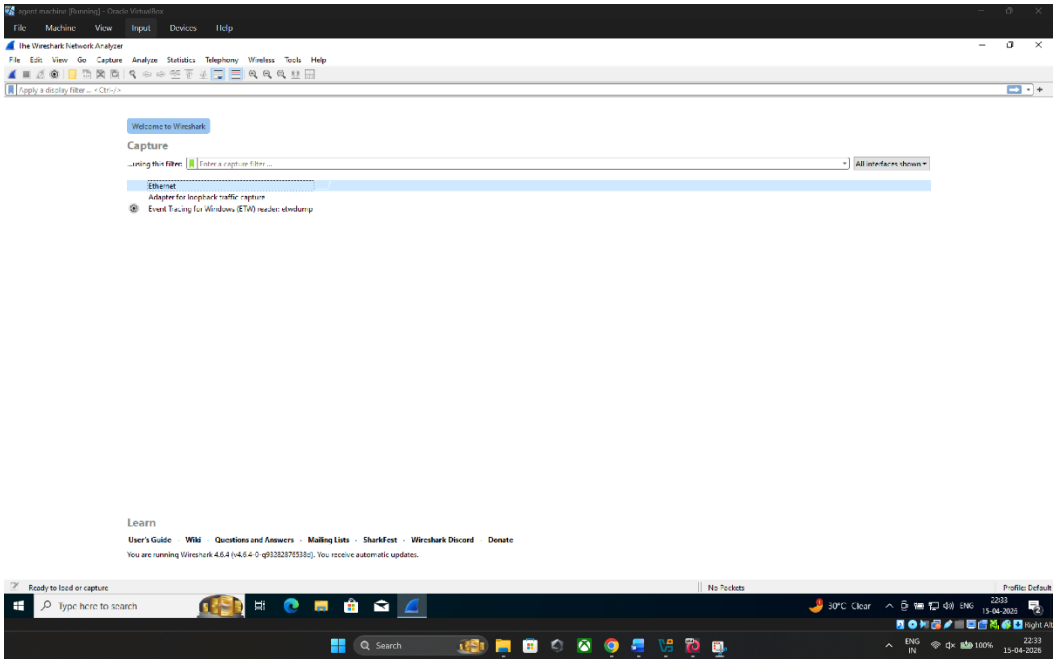
- The target starts Telnet service –



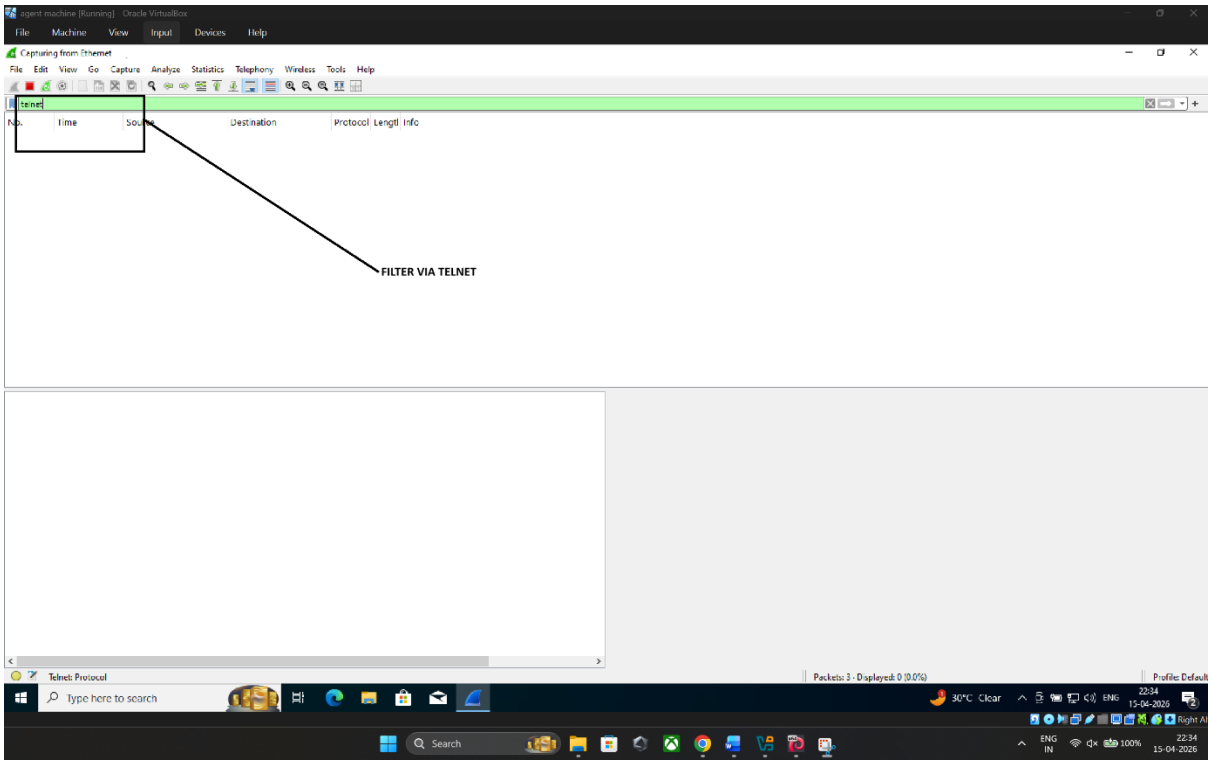
- The target logs on to telnet server



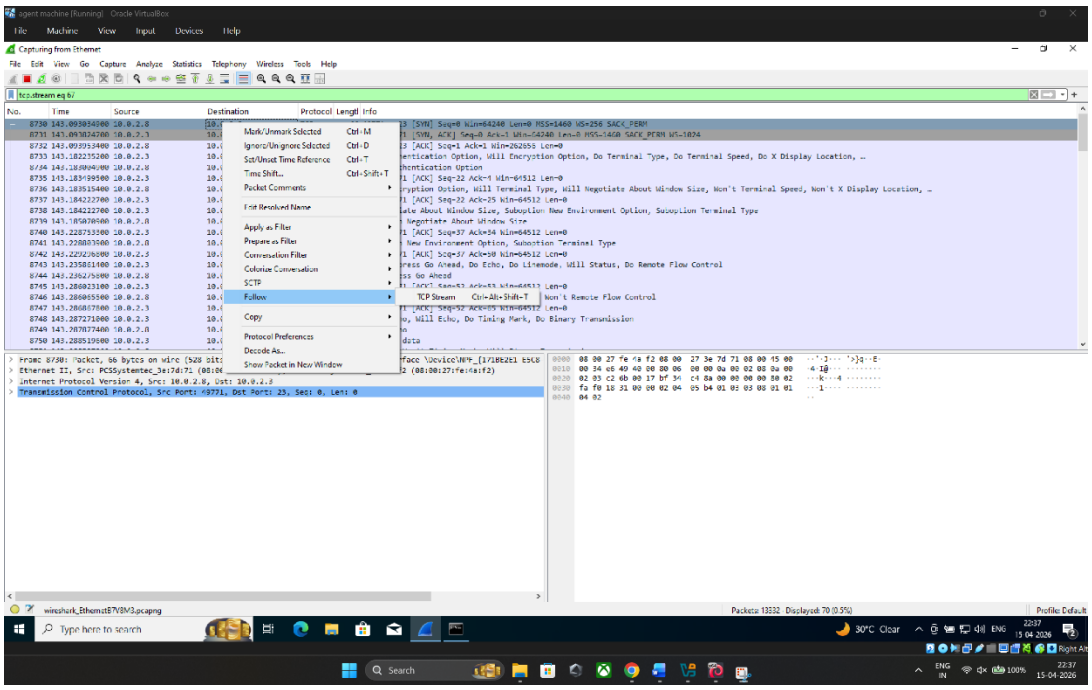
- We open wireshark and select our network adapter



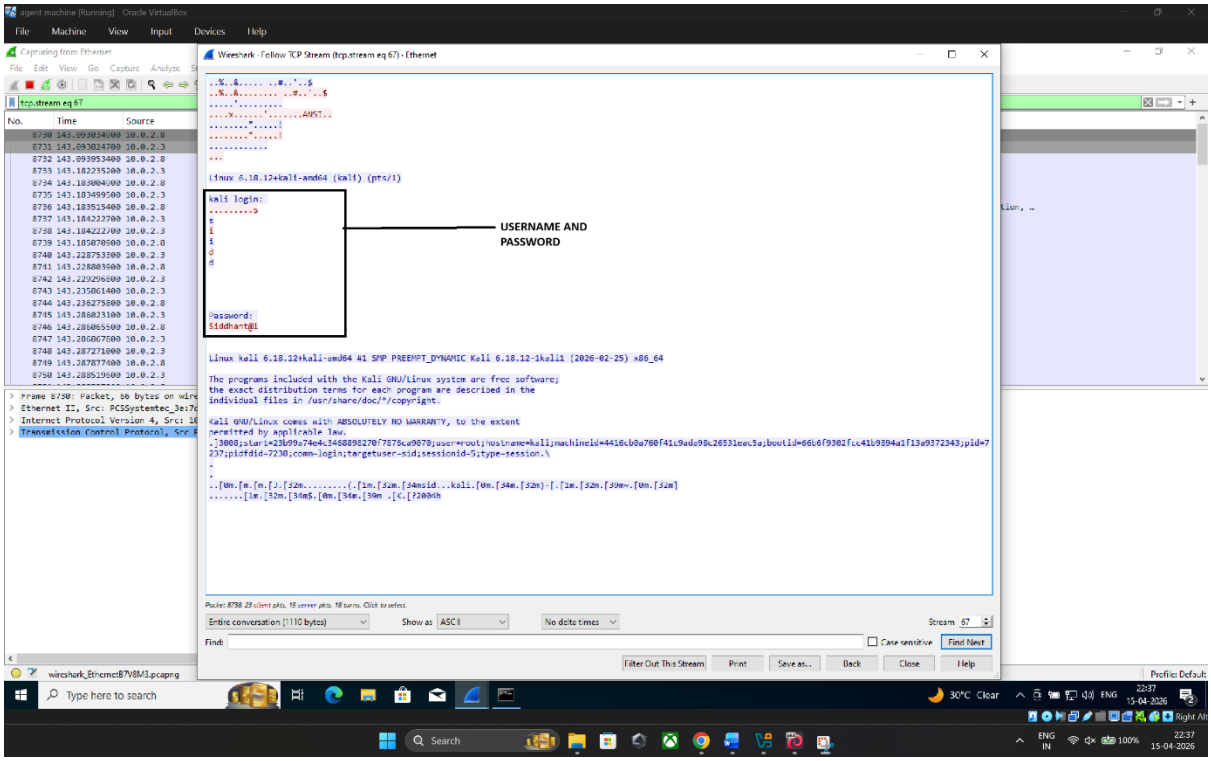
- We filter using Telnet



- We follow TCP Stream on a packet



- We can see the credentials



- If we open packet, we can see source and destination ip aswell

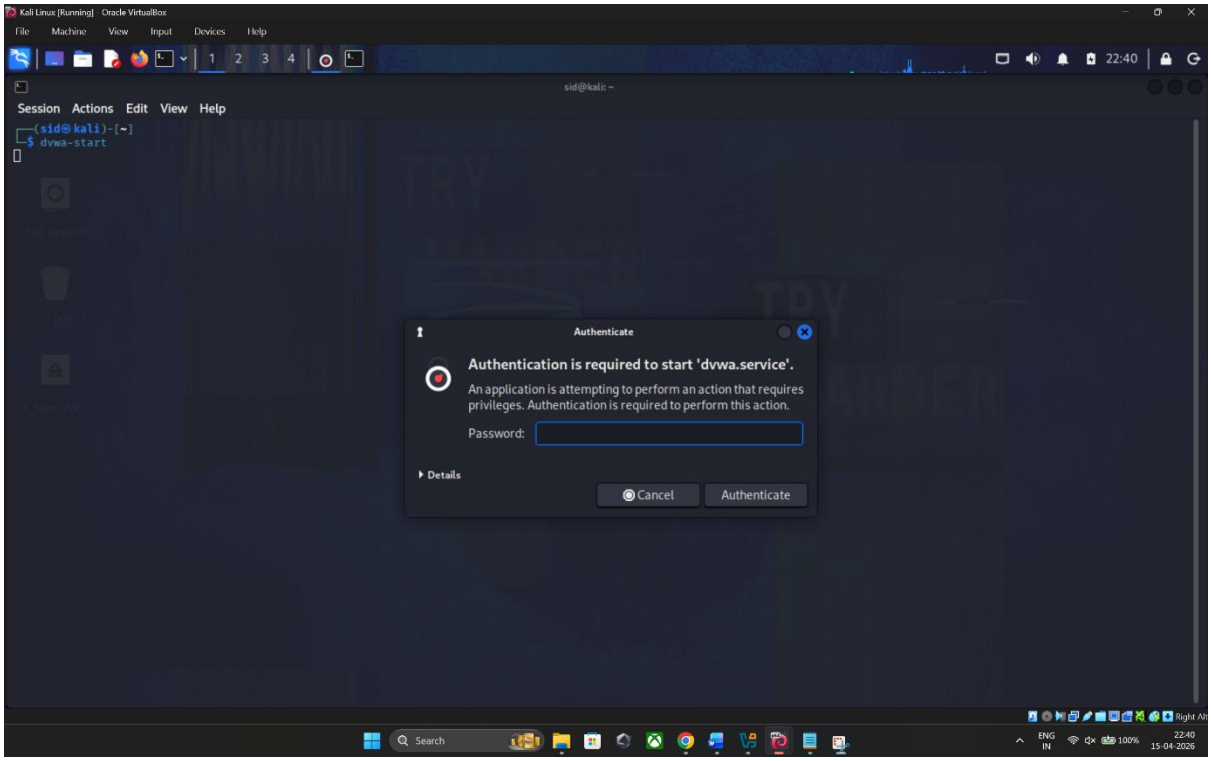
Why encryption absence is dangerous

The absence of encryption is dangerous because Telnet transmits data as clear-text. As seen in the TCP Stream, a passive attacker on the network can reconstruct the session to harvest credentials and monitor all command-line activity, leading to full system compromise."

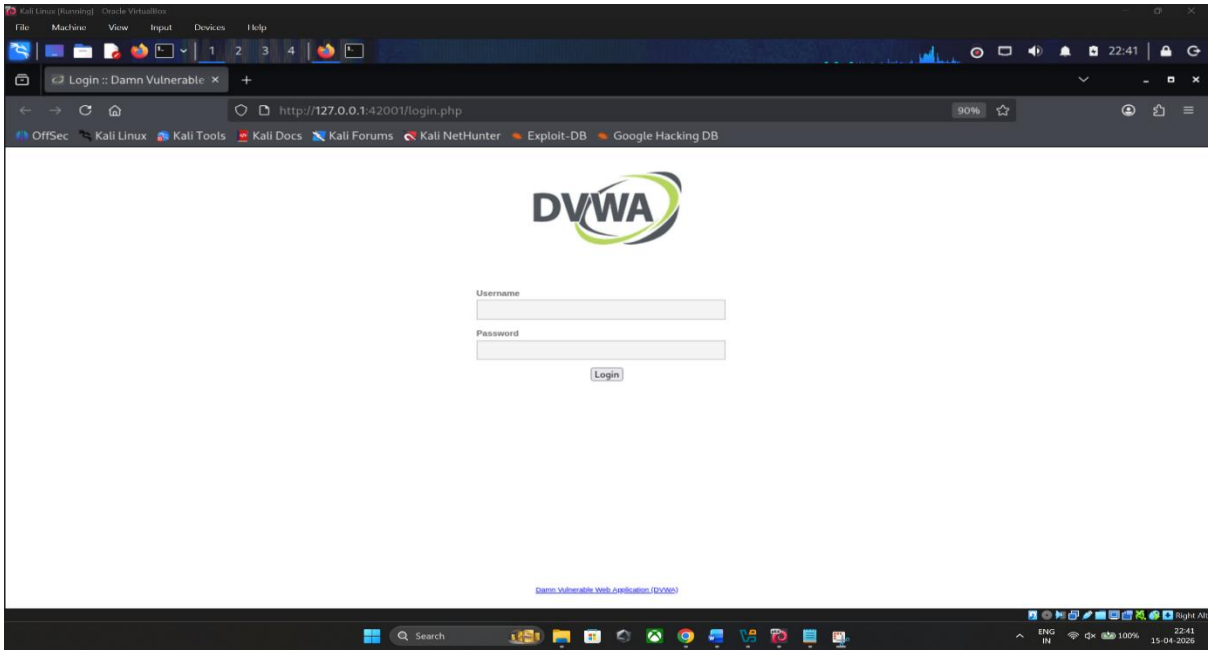
DVWA

- **Capture HTTP POST request and identify submitted credentials.**

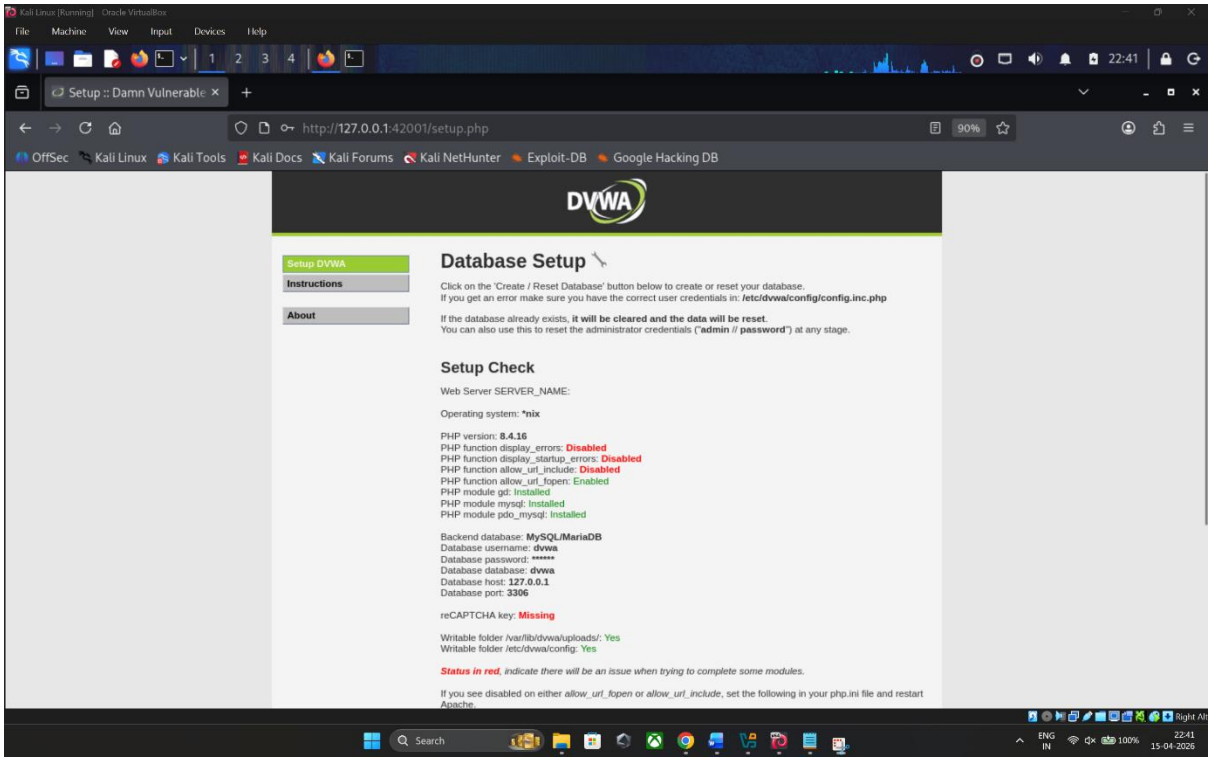
1. The target will start DVWA on his machine



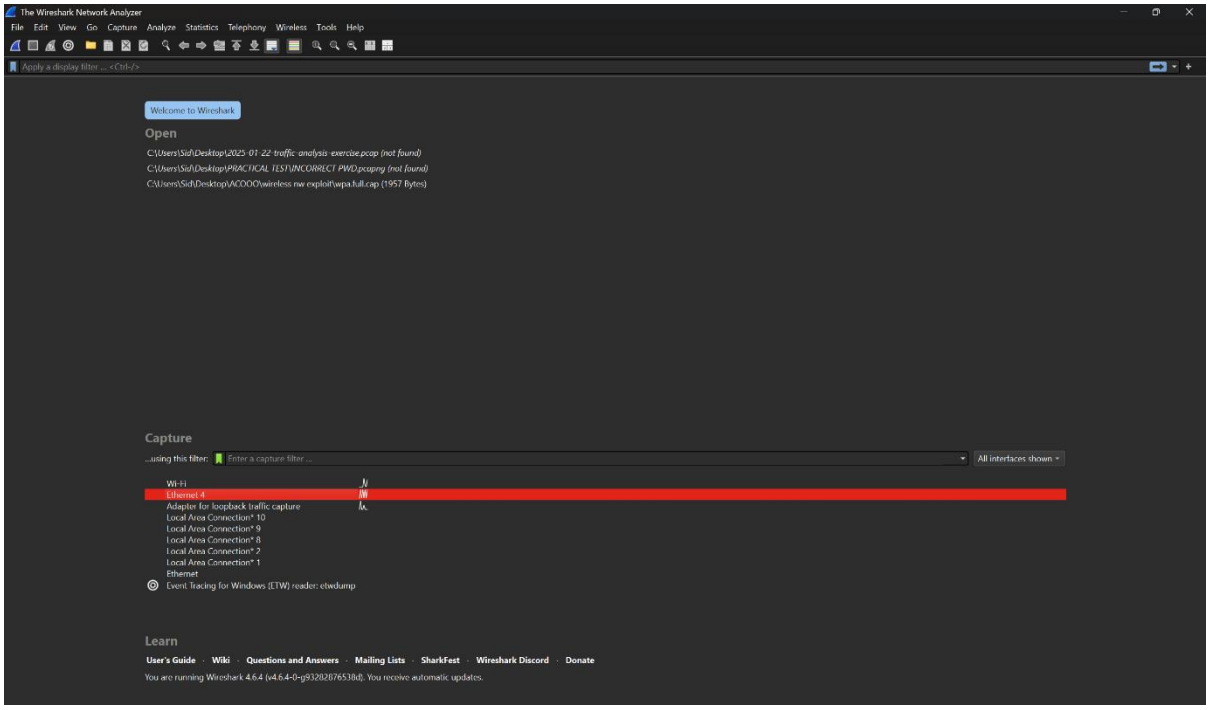
2. Upon starting, DVWA opens up on web browser –



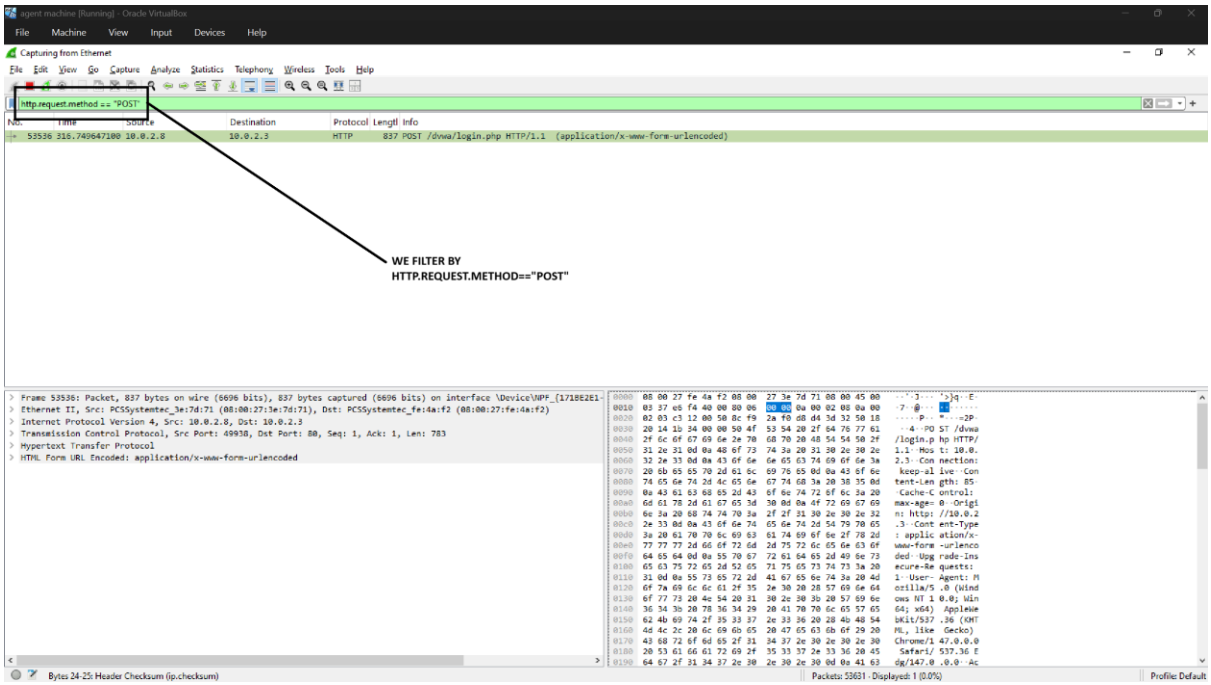
3. Target enters his credentials and logs in –



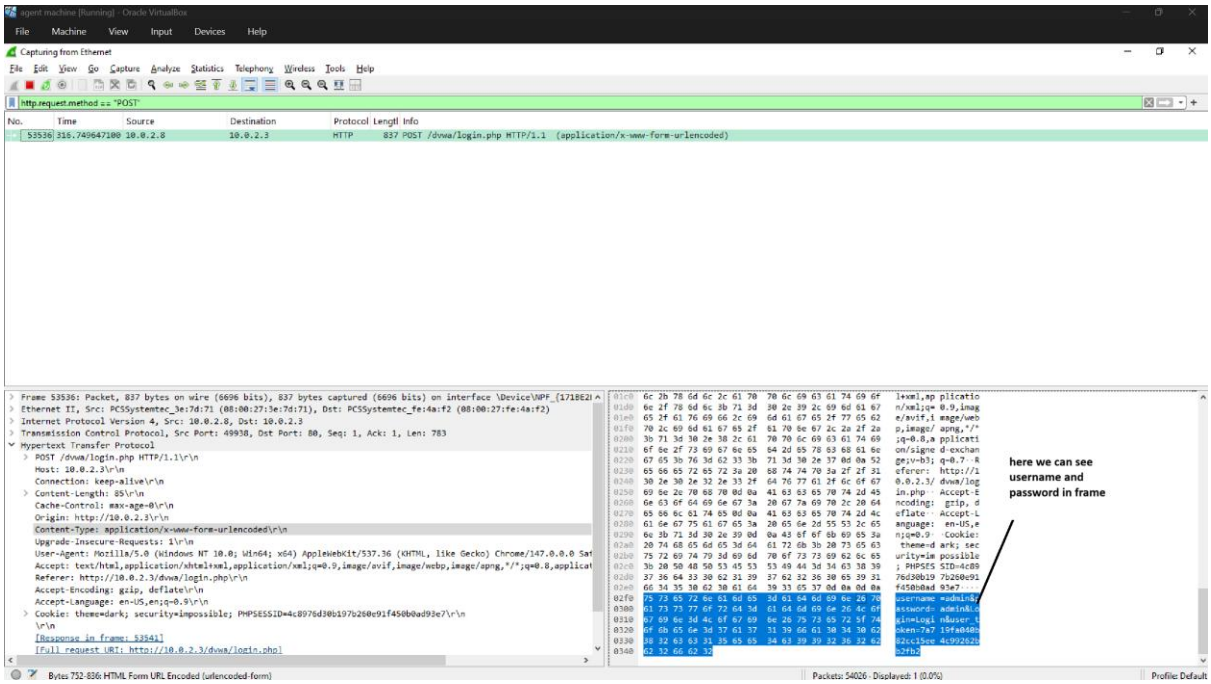
4. At the same time we open wireshark and connect to same network –



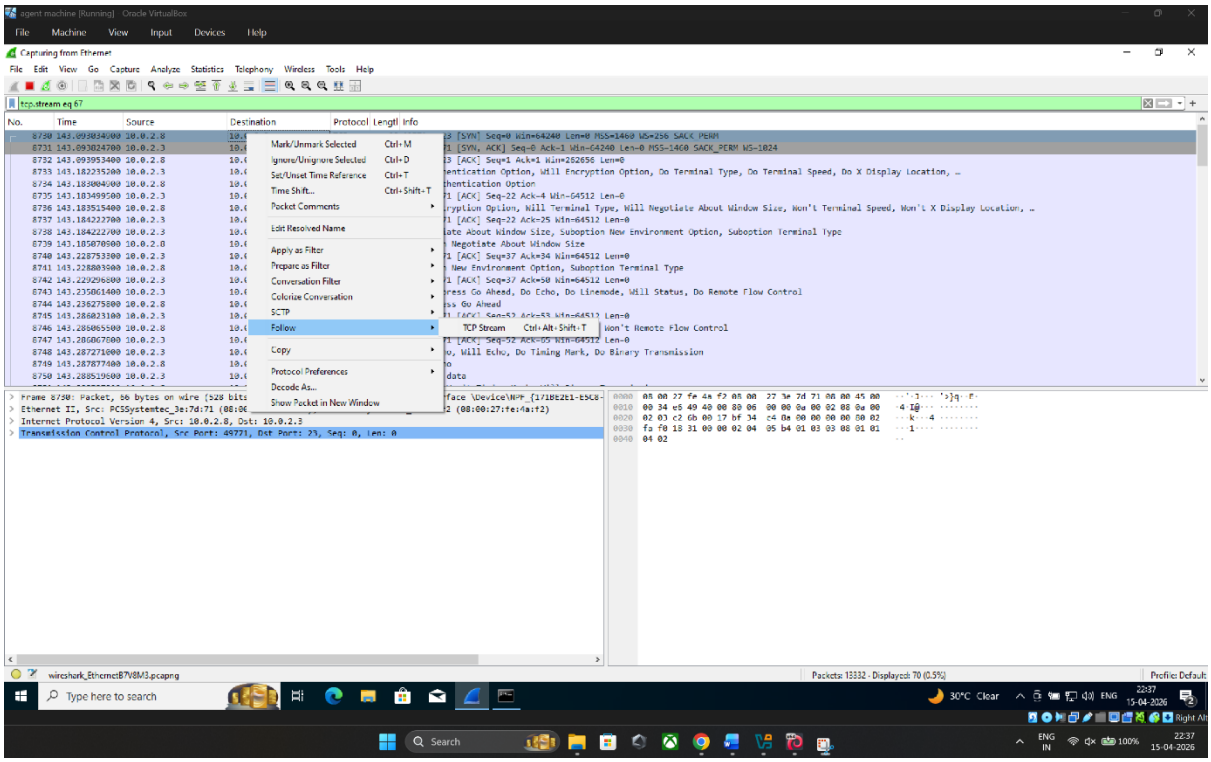
5. Filter Search by POST method –



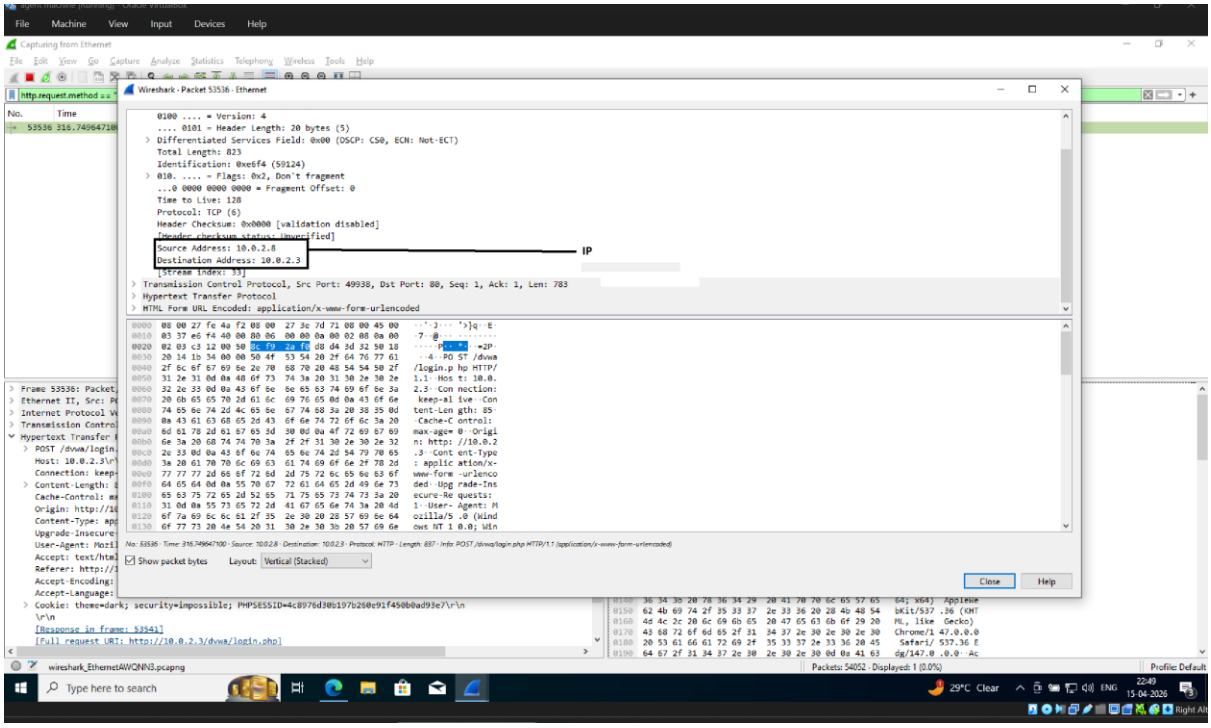
6. We can see password in the frame itself –



7. Can also see credentials in TCP Stream –



8. We can find out Source and Destination IP By opening Packet –



- ## Why encryption absence is dangerous

  HTTP (Port 80) provides no protection for web-based applications. By capturing the **POST request**, I extracted the administrative credentials. In a real-world scenario, an adversary could use this lack of encryption to bypass application security entirely, leading to data exfiltration or unauthorized system control.